

FAST NUCES
Software Engineering

PRISM

Presence Real-time Identification & Security Model

Deliverable #2 - Iteration 1 / Sprint 1

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1. Project Introduction

Project Title: PRISM - Presence Real-time Identification & Security Model

Type: Web Application (React, Node.js, MongoDB)

1.1 Problem Statement

With the shift toward online education, verifying genuine student presence has become difficult. Traditional methods like passwords or login timestamps are easy to abuse through proxy attendance. Many existing systems use facial recognition only, which does not work for niqab-wearing students, or rely on GPS tracking, which raises privacy concerns. There is a need for a secure and inclusive attendance verification system.

1.2 Proposed Solution

PRISM is a web-based attendance verification system that uses facial recognition and fingerprint authentication. It includes fraud prevention through liveness checks and a confidence score. Students can choose their preferred verification method, making it more inclusive. The system runs entirely in the browser and does not use location tracking.

1.3 Technology Stack

- Frontend: React.js with Tailwind CSS
- Backend: Node.js with Express.js
- Database: MongoDB
- AI Service: Python (facial recognition and fingerprint verification)
- Deployment: Vercel (frontend) + Render (backend)

2. Sprint Backlog - Sprint 1

2.1 Module: User Management & Foundation

This sprint covers the core foundation of PRISM - user registration and login, role-based dashboards, and the basic biometric enrollment interface. These are the building blocks for the next sprints.

2.2 Sprint Backlog Table

Sprint Duration: 2 Weeks | Total Estimated Hours: 30 | Sprint Goal: Working login system with role-based dashboards and biometric enrollment interface.

Story ID	Priority	User Story Title	Est. Hrs	Status
US-01	High	User Registration & Authentication	8	To Do
US-02	High	Biometric Enrollment Interface	10	To Do
US-03	High	Instructor Dashboard & Session Init	7	To Do
US-04	Medium	Theme Switching (Light/Dark)	5	To Do

3. User Stories

3.1 US-01 - User Registration & Authentication

Priority: High | Estimated Hours: 8

User Story:

As an administrator, I want to register and log in with role-based access so that the system can identify me and take me to the right dashboard.

Description:

This covers the complete registration and login flow for all users - students, instructors, and admins. It includes form validation, secure login, and taking users to the correct dashboard after logging in. Passwords are stored securely.

Sub User Stories:

- As a student, I want to register with my university email so I can access the system.
- As an instructor, I want to log in with my credentials so I am taken to the instructor dashboard.
- As an admin, I want to manage user accounts so I can control who has access.
- As a user, I want to see clear error messages when my login fails.

Acceptance Criteria:

- Registration form accepts full name, email, password, and role.
- Password must be at least 8 characters with one uppercase letter and one digit.
- Users are redirected to their role-specific dashboard after login.
- Incorrect login shows an error without revealing which field is wrong.

3.2 US-02 - Biometric Enrollment Interface

Priority: High | Estimated Hours: 10

User Story:

As a student, I want to enroll my biometric data (face and/or fingerprint) so that I can verify my attendance without using shared credentials.

Description:

This covers the interface where students enroll their biometric data. The process guides the user step by step, captures data through webcam or fingerprint scanner, sends it to the backend for secure storage, and confirms when enrollment is successful. Students are never forced into one method only.

Sub User Stories:

- As a student, I want to choose between face, fingerprint, or both enrollment modes.
- As a student, I want step-by-step guidance during the enrollment process.
- As a student, I want confirmation that my biometric data was saved successfully.
- As a student, I want to re-enroll if my initial attempt fails quality checks.

Acceptance Criteria:

- Enrollment screen is accessible from the student dashboard after first login.
- Users can choose face-only, fingerprint-only, or both.
- Webcam feed is shown live during face enrollment with alignment guides.
- A confirmation message appears when enrollment is complete.

3.3 US-03 - Instructor Dashboard & Session Initialisation

Priority: High | Estimated Hours: 7

User Story:

As an instructor, I want to create and manage attendance sessions so that students can mark attendance within a set time window.

Description:

This covers the instructor dashboard where instructors can view their courses, create attendance sessions for specific classes, and see real-time attendance progress. Sessions have a fixed start and end time, and students can only mark attendance during that window.

Sub User Stories:

- As an instructor, I want to see all my assigned courses from my dashboard.
- As an instructor, I want to create a new attendance session with a start and end time.
- As an instructor, I want to see live progress of who has marked attendance.
- As an instructor, I want to end a session early if needed.

Acceptance Criteria:

- Dashboard shows all courses assigned to the logged-in instructor.
- Instructor can create a session by selecting a course, date, and time window.
- Active session shows a live count of students who have verified.
- Students outside the session window see a 'Session Closed' message.
- Instructor can close a session manually at any time.

3.4 US-04 - Theme Switching (Light / Dark)

Priority: Medium | Estimated Hours: 5

User Story:

As a user, I want to switch between light and dark themes so I can personalise how the app looks.

Description:

This covers a theme switching feature that lets users toggle between light and dark mode. The selected theme should be saved and applied across all screens without needing a page reload.

Sub User Stories:

- As a user, I want to select between light and dark themes.

- As a user, I want my theme choice to be remembered between sessions.
- As a user, I want the theme to apply consistently across the whole app.

Acceptance Criteria:

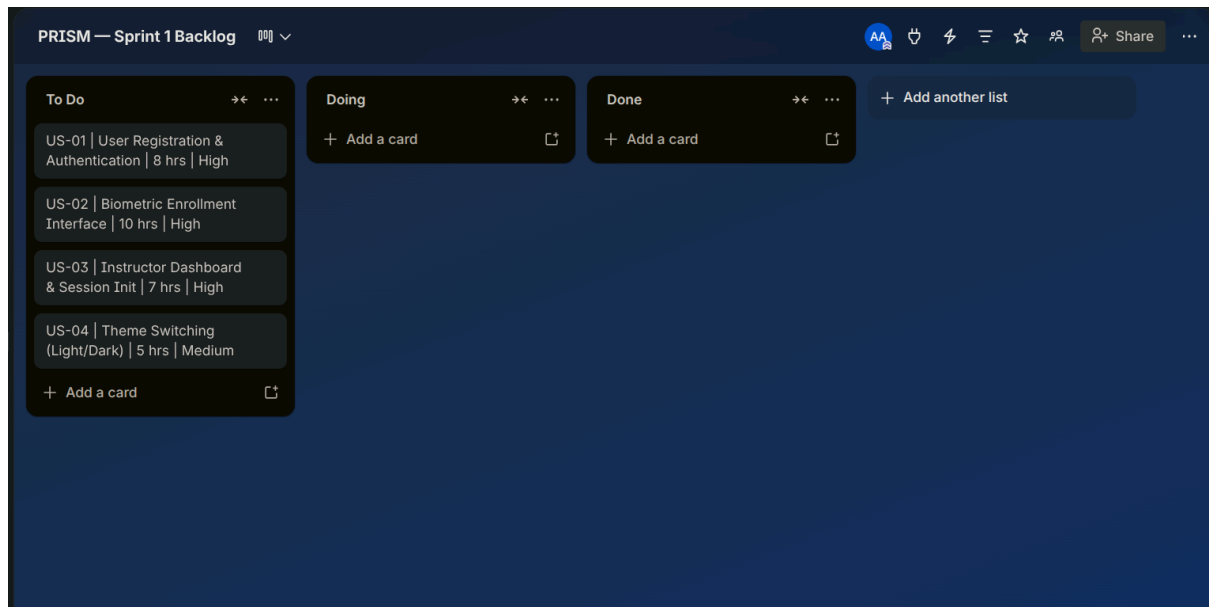
- Theme toggle is accessible from the Settings screen.
- Both light and dark options are available to all users.
- Selected theme is saved and persists across sessions.
- All UI components reflect the active theme.
- Theme change applies without a full page reload.

4. Scrum Board (Trello)

The sprint backlog was uploaded to Trello and maintained throughout the sprint. Three snapshots were taken at different stages of the sprint.

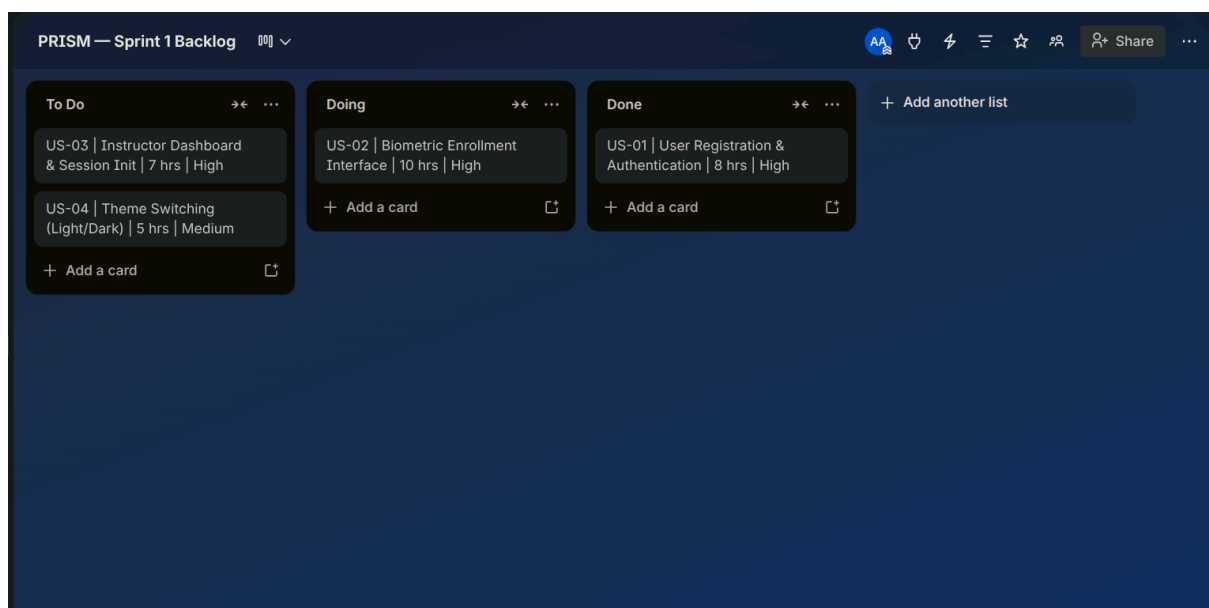
Snapshot 1 - Sprint Start (Day 1)

All four user stories uploaded to the 'To Do' column. Sprint started with 30 estimated hours.



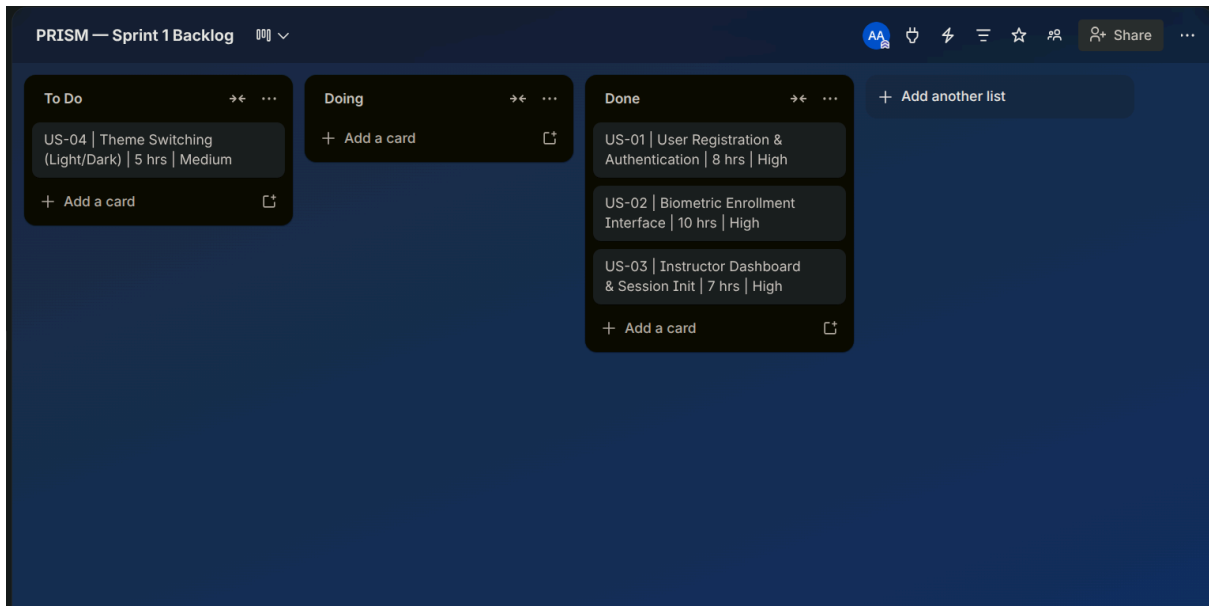
Snapshot 2 - Mid-Sprint (Day 7)

US-01 completed and moved to 'Done'. US-02 moved to 'In Progress'. US-03 initiated. Around 50% of sprint work done.



Snapshot 3 - Sprint End (Day 14)

US-01, US-02, and US-03 moved to 'Done'. US-04 (Theme Switching) marked as a Leftover and moved to Sprint 2.

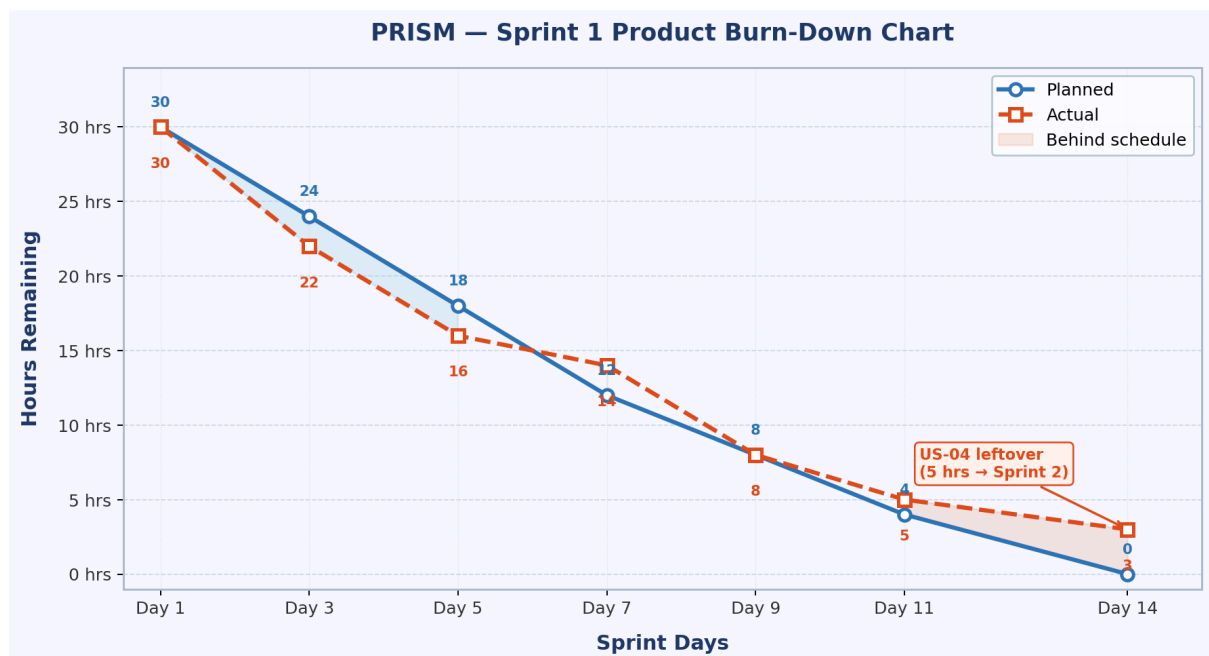


5. Product Burn-Down Chart - Sprint 1

The table below shows planned vs. actual hours remaining across the 14-day sprint.

Day	Planned Hours Remaining	Actual Hours Remaining	Notes
Day 1	30	30	Sprint start – all stories in backlog
Day 3	24	22	US-01 auth in progress
Day 5	18	16	US-01 completed; US-02 started
Day 7	12	14	US-02 enrollment UI in progress
Day 9	8	8	US-03 session management in progress
Day 11	4	5	US-03 completed; US-04 started
Day 14	0	3	US-04 moved to Sprint 2 as leftover

Note: US-04 (Theme Switching, 5 hours) was moved to Sprint 2 as we focused on finishing the core authentication and biometric stories first.



6. Non-Functional Requirements

The following basic non-functional requirements apply to the PRISM system:

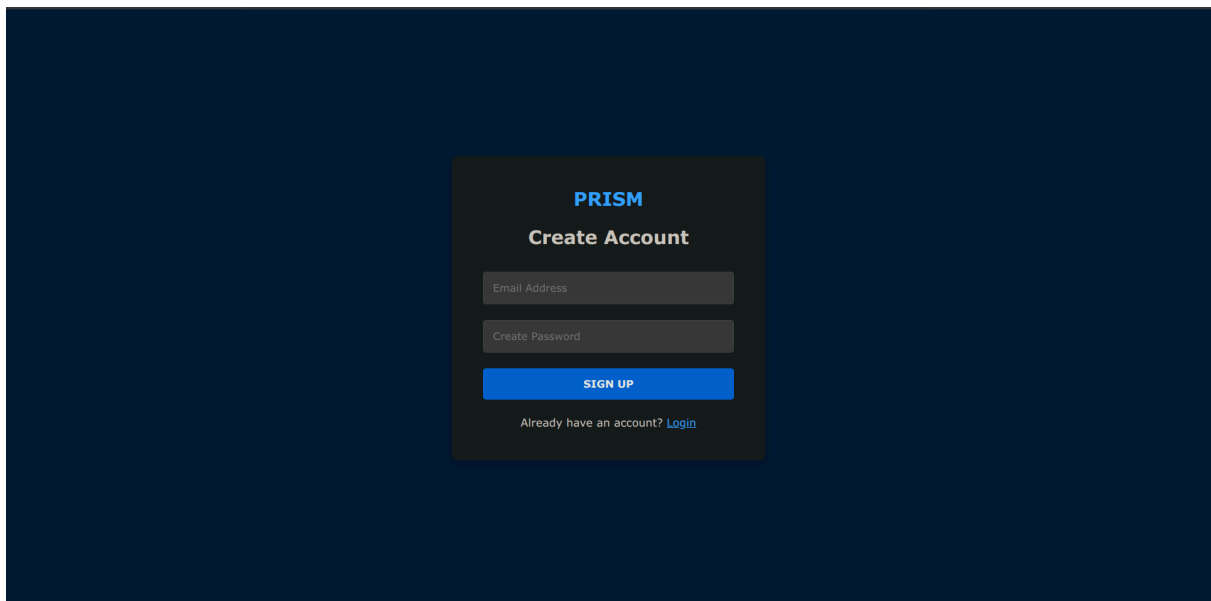
- Security: All user passwords are stored securely using hashing. Biometric data is stored as encrypted templates only — no raw images are saved anywhere.
- Performance: Pages should load within 2 seconds on a standard connection. Login and registration responses should be fast under normal usage.
- Usability: The app should work on both desktop and mobile browsers. All main actions should be reachable within a few clicks from the dashboard.
- Reliability: The system should aim for high uptime. If the backend goes down, users should see a clear error message rather than a blank screen.

7. Implementation Screenshots

The following screenshots show the working implementation at the end of Sprint 1.

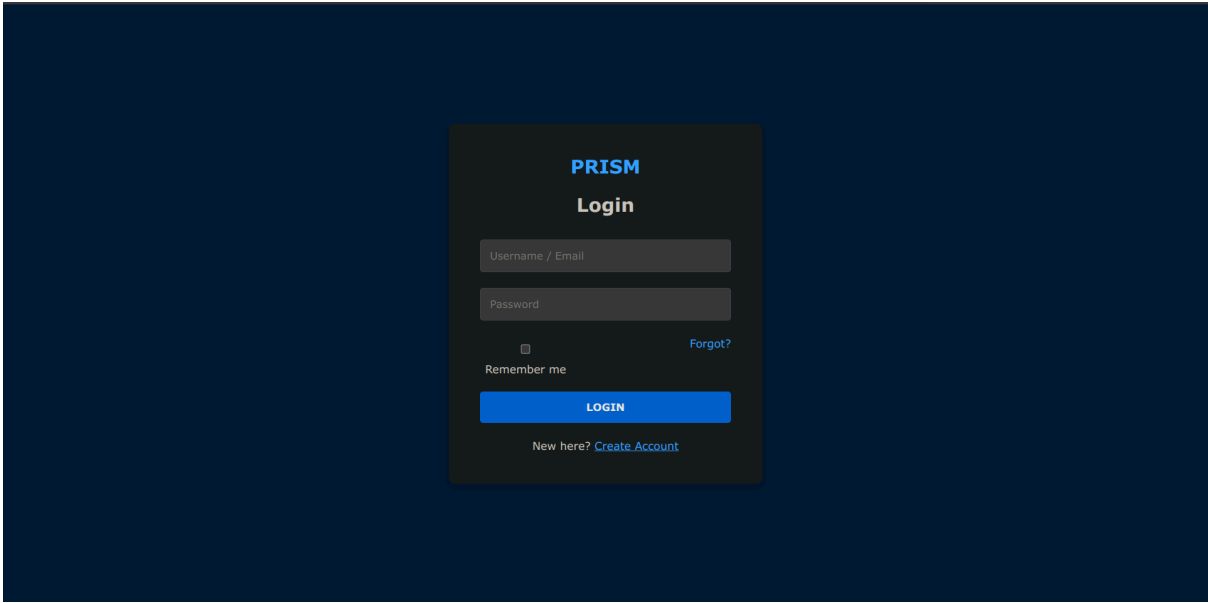
Screenshot 1 - Sign Up Page

The user registration form with email, password, and role fields.

A screenshot of the PRISM 'Create Account' sign-up page. The page has a dark blue background. In the center, there is a dark gray rectangular box containing the form. At the top of this box, the word 'PRISM' is written in blue, and 'Create Account' is written in white. Below the title, there are two input fields: 'Email Address' and 'Create Password', both with light gray text. Under these fields is a bright blue button with the text 'SIGN UP' in white. At the bottom of the box, there is a link that says 'Already have an account? Login' in a small, light blue font.

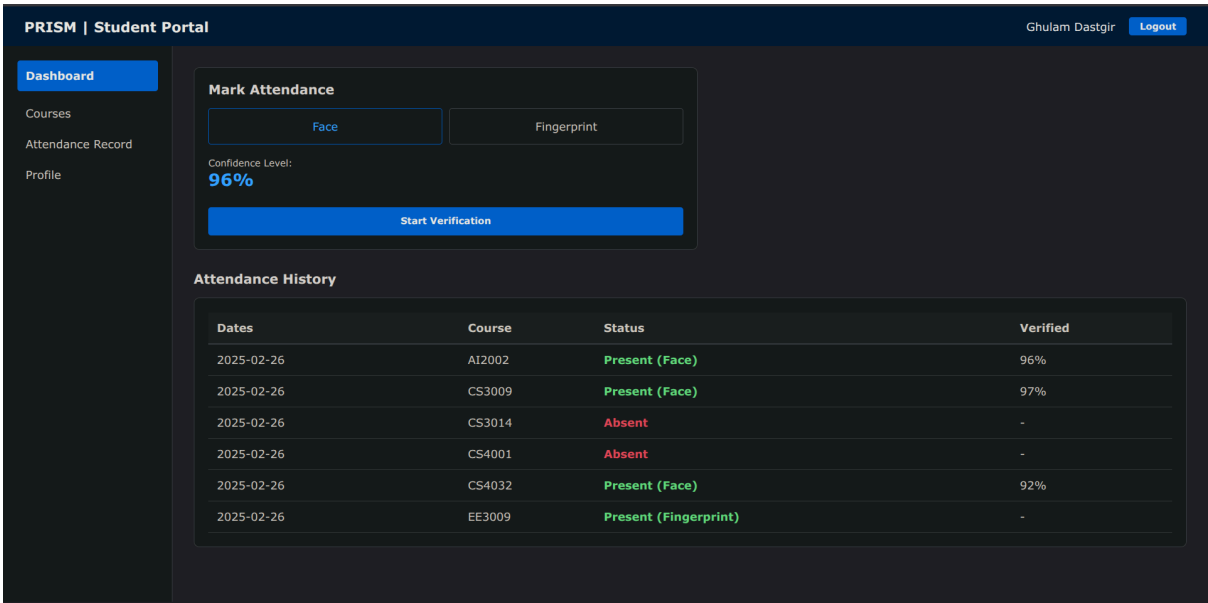
Screenshot 2 - Login Page

The login screen with email and password fields.



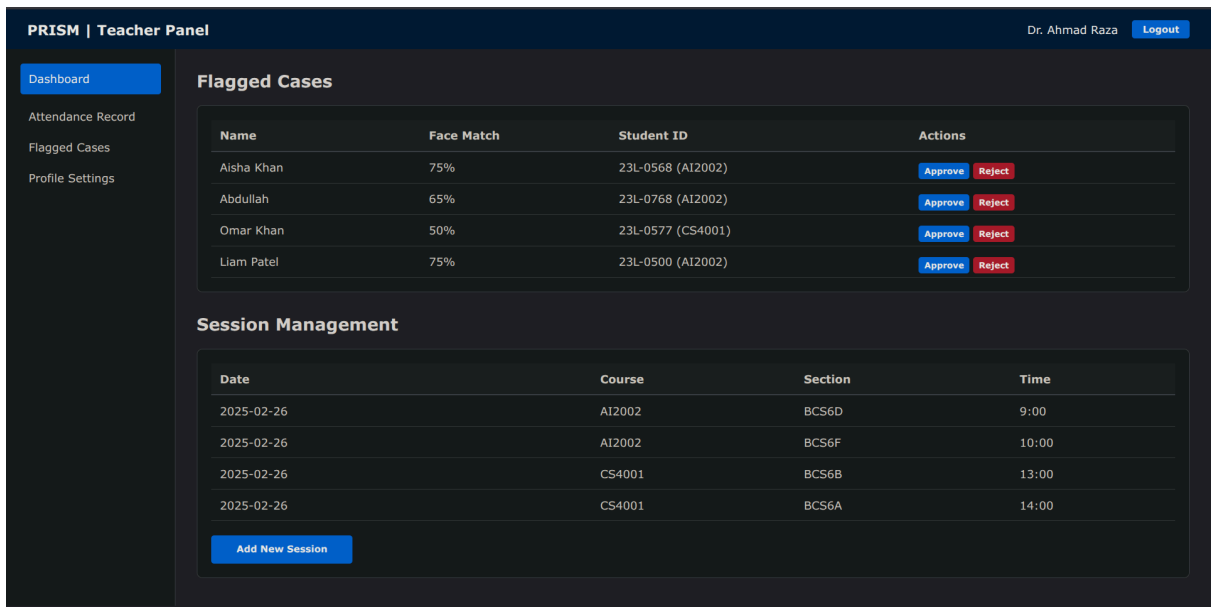
Screenshot 3 - Student Dashboard

Student dashboard showing the attendance marking section, biometric options, and attendance history.



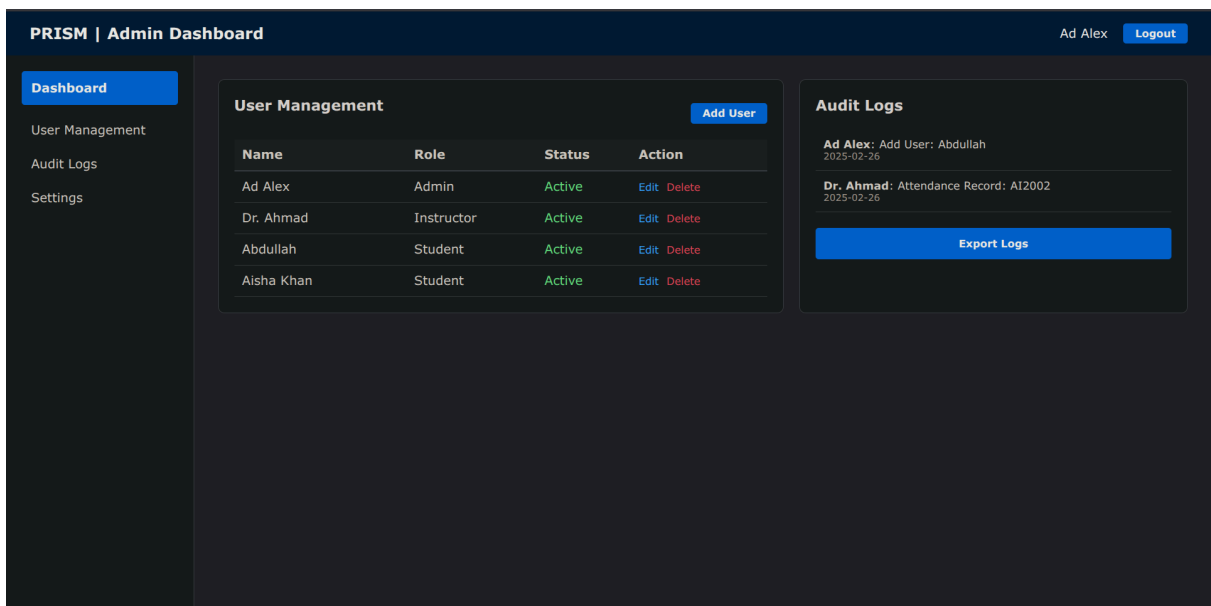
Screenshot 4 - Teacher Dashboard

Instructor dashboard showing the course list, session management, and flagged cases.



Screenshot 5 -Admin Dashboard

Admin panel showing the user management list and audit logs.



8. Work Division

This is a solo project. All work was done by one team member.

Member	Role	Responsibilities
Areeb Ahmed	Project Lead & Developer	System design, full frontend and backend development, AI integration, testing and documentation

Note: All development, testing, documentation, and project management was done by Areeb Ahmed (23L-0783). This is a solo project approved by the course instructor.